

Differential Equations (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

49

TOTAL MARKS

81 M

TEST DURATION

147 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	27	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	16	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q52 • ID: JAM_2021_Q52)

NAT

+2 M

Neg: Nil (0)

Q. 52 The least possible value of k , accurate up to two decimal places, for which the following problem

$$y''(t) + 2y'(t) + ky(t) = 0, t \in \mathbb{R},$$
$$y(0) = 0, y(1) = 0, y(1/2) = 1,$$

has a solution is ____.

Question 2 (JAM 2021 • Q48 • ID: JAM_2021_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $y : (\frac{9}{10}, 3) \rightarrow \mathbb{R}$ be a differentiable function satisfying

$$(x - 2y)\frac{dy}{dx} + (2x + y) = 0, \quad x \in \left(\frac{9}{10}, 3\right), \quad \text{and } y(1) = 1.$$

Then $y(2)$ equals ____.

Question 3 (JAM 2021 • Q25 • ID: JAM_2021_Q25)

MCQ

+2 M

Neg: -0.67

JAM 2021MATHEMATICS - MA

Q. 25 Let y be a twice differentiable function on $\mathbb{R}$ satisfying

$$y''(x) = 2 + e^{-|x|}, \quad x \in \mathbb{R},$$
$$y(0) = -1, \quad y'(0) = 0.$$

Then

- (A) $y = 0$ has exactly one root.
- (B) $y = 0$ has exactly two roots.
- (C) $y = 0$ has more than two roots.
- (D) there exists an $x_0 \in \mathbb{R}$ such that $y(x_0) \geq y(x)$ for all $x \in \mathbb{R}$.

Question 4 (JAM 2021 • Q19 • ID: JAM_2021_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Consider the family of curves $x^2 - y^2 = ky$ with parameter $k \in \mathbb{R}$. The equation of the orthogonal trajectory to this family passing through $(1, 1)$ is given by

(A) $x^3 + 3xy^2 = 4$.

(B) $x^2 + 2xy = 3$.

(C) $y^2 + 2x^2y = 3$.

(D) $x^3 + 2xy^2 = 3$.

Question 5 (JAM 2021 • Q12 • ID: JAM_2021_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Let y be the solution of

$$(1+x)y''(x) + y'(x) - \frac{1}{1+x}y(x) = 0, \quad x \in (-1, \infty),$$
$$y(0) = 1, \quad y'(0) = 0.$$

Then

(A) y is bounded on $(0, \infty)$.(B) y is bounded on $(-1, 0]$.(C) $y(x) \geq 2$ on $(-1, \infty)$.(D) y attains its minimum at $x = 0$.

Question 6 (JAM 2021 • Q2 • ID: JAM_2021_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $P(x) > 0$ for all $x \in \mathbb{R}$. Let y be a twice differentiable function on $\mathbb{R}$ satisfying $y''(x) + P(x)y'(x) - y(x) = 0$ for all $x \in \mathbb{R}$. Suppose that there exist two real numbers a, b ($a < b$) such that $y(a) = y(b) = 0$. Then

(A) $y(x) = 0$ for all $x \in [a, b]$.(B) $y(x) > 0$ for all $x \in (a, b)$.(C) $y(x) < 0$ for all $x \in (a, b)$.(D) $y(x)$ changes sign on (a, b) .

Question 7 (JAM 2021 • Q1 • ID: JAM_2021_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 Let $0 < \alpha < 1$ be a real number. The number of differentiable functions $y : [0, 1] \rightarrow [0, \infty)$, having continuous derivative on $[0, 1]$ and satisfying

$$\begin{aligned} y'(t) &= (y(t))^\alpha, \quad t \in [0, 1], \\ y(0) &= 0, \end{aligned}$$

is

(A) exactly one.

(C) finite but more than two.

(B) exactly two.

(D) infinite.

Question 8 (JAM 2020 • Q57 • ID: JAM_2020_Q57)

NAT

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 57 If $x^2 + xy^2 = c$, where $c \in \mathbb{R}$, is the general solution of the exact differential equation

$$M(x, y) dx + 2xy dy = 0,$$

then $M(1, 1)$ is _____.

$$\begin{bmatrix} 9 & 2 & 7 & 1 \end{bmatrix}$$

Question 9 (JAM 2020 • Q56 • ID: JAM_2020_Q56)

NAT

+2 M

Neg: Nil (0)

Q. 56 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f'(x) = f(x)$ for all x . Suppose that $f(\alpha x)$ and $f(\beta x)$ are two non-zero solutions of the differential equation

$$4 \frac{d^2 y}{dx^2} - p \frac{dy}{dx} + 3y = 0$$

satisfying

$$f(\alpha x)f(\beta x) = f(2x) \quad \text{and} \quad f(\alpha x)f(-\beta x) = f(x).$$

Then, the value of p is _____.

MA

16 / 17

Question 10 (JAM 2020 • Q48 • ID: JAM_2020_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $f(x, y) = 0$ be a solution of the homogeneous differential equation

$$(2x + 5y)dx - (x + 3y)dy = 0.$$

If $f(x + \alpha, y - 3) = 0$ is a solution of the differential equation

$$(2x + 5y - 1)dx + (2 - x - 3y)dy = 0,$$

then the value of α is _____.

Question 11 (JAM 2020 • Q46 • ID: JAM_2020_Q46)

NAT

+1 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 46 Consider the differential equation

$$\frac{dy}{dx} + 10y = f(x), \quad x > 0,$$

where $f(x)$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = 1$. Then the value of

$$\lim_{x \rightarrow \infty} y(x)$$

is _____.

Question 12 (JAM 2020 • Q35 • ID: JAM_2020_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Let $L[y] = x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} + qy$, where p, q are real constants. Let $y_1(x)$ and $y_2(x)$ be two solutions of $L[y] = 0, x > 0$, that satisfy $y_1(x_0) = 1, y_1'(x_0) = 0, y_2(x_0) = 0$ and $y_2'(x_0) = 1$ for some $x_0 > 0$. Then,

- (A) $y_1(x)$ is not a constant multiple of $y_2(x)$
- (B) $y_1(x)$ is a constant multiple of $y_2(x)$
- (C) $1, \ln x$ are solutions of $L[y] = 0$ when $p = 1, q = 0$
- (D) $x, \ln x$ are solutions of $L[y] = 0$ when $p + q \neq 0$

Question 13 (JAM 2020 • Q22 • ID: JAM_2020_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 A solution of the differential equation $2x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} - y = 0, x > 0$ that passes through the point $(1, 1)$ is

(A) $y = \frac{1}{x}$

(B) $y = \frac{1}{x^2}$

(C) $y = \frac{1}{\sqrt{x}}$

(D) $y = \frac{1}{x^{3/2}}$

Question 14 (JAM 2020 • Q21 • ID: JAM_2020_Q21)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 21 Consider the differential equation $L[y] = (y - y^2)dx + xdy = 0$. The function $f(x, y)$ is said to be an integrating factor of the equation if $f(x, y)L[y] = 0$ becomes exact.

If $f(x, y) = \frac{1}{x^2 y^2}$, then

(A) f is an integrating factor and $y = 1 - kxy, k \in \mathbb{R}$ is NOT its general solution

(B) f is an integrating factor and $y = -1 + kxy, k \in \mathbb{R}$ is its general solution

(C) f is an integrating factor and $y = -1 + kxy, k \in \mathbb{R}$ is NOT its general solution

(D) f is NOT an integrating factor and $y = 1 + kxy, k \in \mathbb{R}$ is its general solution

Question 15 (JAM 2020 • Q7 • ID: JAM_2020_Q7)

MCQ

+1 M

Neg: -0.33

Q. 7 If $u = x^3$ and $v = y^2$ transform the differential equation $3x^5 dx - y(y^2 - x^3)dy = 0$ to $\frac{dv}{du} = \frac{\alpha u}{2(u - v)}$, then α is

(A) 4

(B) 2

(C) -2

(D) -4

Question 16 (JAM 2019 • Q60 • ID: JAM_2019_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x)$ is the solution of the initial value problem

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0, \quad y(0) = 2, \quad \frac{dy}{dx}(0) = 0,$$

then $y(\ln 2)$ is (round off to 2 decimal places) equal to _____

END OF THE QUESTION PAPER

Question 17 (JAM 2019 • Q59 • ID: JAM_2019_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let $y(x) = xv(x)$ be a solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 3y = 0.$$

If $v(0) = 0$ and $v(1) = 1$, then $v(-2)$ is equal to _____

MA

12/13

Question 18 (JAM 2019 • Q21 • ID: JAM_2019_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 Let x , $x + e^x$ and $1 + x + e^x$ be solutions of a linear second order ordinary differential equation with constant coefficients. If $y(x)$ is the solution of the same equation satisfying $y(0) = 3$ and $y'(0) = 4$, then $y(1)$ is equal to(A) $e + 1$ (B) $2e + 3$ (C) $3e + 2$ (D) $3e + 1$

Question 19 (JAM 2019 • Q20 • ID: JAM_2019_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let S be the family of orthogonal trajectories of the family of curves

$$2x^2 + y^2 = k, \text{ for } k \in \mathbb{R} \text{ and } k > 0.$$

If $C \in S$ and C passes through the point $(1, 2)$, then C also passes through(A) $(4, -\sqrt{2})$ (B) $(2, -4)$ (C) $(2, 2\sqrt{2})$ (D) $(4, 2\sqrt{2})$

Question 20 (JAM 2019 • Q7 • ID: JAM_2019_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 If $y(x) = \lambda e^{2x} + e^{\beta x}$, $\beta \neq 2$, is a solution of the differential equation

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

satisfying $\frac{dy}{dx}(0) = 5$, then $y(0)$ is equal to

(A) 1

(B) 4

(C) 5

(D) 9

Question 21 (JAM 2019 • Q6 • ID: JAM_2019_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 If $x^h y^k$ is an integrating factor of the differential equation

$$y(1 + xy) dx + x(1 - xy) dy = 0,$$

then the ordered pair (h, k) is equal to

- (A) $(-2, -2)$ (B) $(-2, -1)$ (C) $(-1, -2)$ (D) $(-1, -1)$

Question 22 (JAM 2018 • Q60 • ID: JAM_2018_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x) = v(x) \sec x$ is the solution of $y'' - (2 \tan x) y' + 5y = 0$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, satisfying

$y(0) = 0$ and $y'(0) = \sqrt{6}$, then $v\left(\frac{\pi}{6\sqrt{6}}\right)$ is _____ (correct up to two decimal places).

END OF THE QUESTION PAPER

MA

12/12

Question 23 (JAM 2018 • Q59 • ID: JAM_2018_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 If $x(t)$ is the solution to the differential equation $\frac{dx}{dt} = x^2 t^3 + xt$, for $t > 0$, satisfying $x(0) = 1$,

then the value of $x(\sqrt{2})$ is _____ (correct up to two decimal places).

Question 24 (JAM 2018 • Q46 • ID: JAM_2018_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $f: [0, \infty) \rightarrow [0, \infty)$ be continuous on $[0, \infty)$ and differentiable on $(0, \infty)$. If

$$f(x) = \int_0^x \sqrt{f(t)} dt, \text{ then } f(6) = \underline{\hspace{2cm}}$$

MA

10/12

Question 25 (JAM 2018 • Q32 • ID: JAM_2018_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The solution(s) of the differential equation $\frac{dy}{dx} = (\sin 2x) y^{1/3}$ satisfying $y(0) = 0$ is (are)

(A) $y(x) = 0$

(B) $y(x) = -\sqrt{\frac{8}{27}} \sin^3 x$

(C) $y(x) = \sqrt{\frac{8}{27}} \sin^3 x$

(D) $y(x) = \sqrt{\frac{8}{27}} \cos^3 x$

Question 26 (JAM 2018 • Q24 • ID: JAM_2018_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 A particular integral of the differential equation $y'' + 3y' + 2y = e^x$ is

(A) $e^{e^x} e^{-x}$

(B) $e^{e^x} e^{-2x}$

(C) $e^{e^x} e^{2x}$

(D) $e^{e^x} e^x$

Question 27 (JAM 2018 • Q23 • ID: JAM_2018_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 An integrating factor of the differential equation $\left(y + \frac{1}{3} y^3 + \frac{1}{2} x^2\right) dx + \frac{1}{4} (x + x y^2) dy = 0$ is

(A) x^2

(B) $3 \log_e x$

(C) x^3

(D) $2 \log_e x$

Question 28 (JAM 2018 • Q22 • ID: JAM_2018_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $y(x)$ be the solution of the differential equation $\frac{dy}{dx} + y = f(x)$, for $x \geq 0$, $y(0) = 0$, where

$$f(x) = \begin{cases} 2, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}. \quad \text{Then } y(x) =$$

- (A) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(e - 1)e^{-x}$ when $x \geq 1$
 (B) $2(1 - e^{-x})$ when $0 \leq x < 1$ and 0 when $x \geq 1$
 (C) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(1 - e^{-1})e^{-x}$ when $x \geq 1$
 (D) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2e^{1-x}$ when $x \geq 1$

Question 29 (JAM 2018 • Q8 • ID: JAM_2018_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 In $\mathbb{R}^2$, the family of trajectories orthogonal to the family of asteroids $x^{2/3} + y^{2/3} = a^{2/3}$ is given by

- (A) $x^{4/3} + y^{4/3} = c^{4/3}$ (B) $x^{4/3} - y^{4/3} = c^{4/3}$
 (C) $x^{5/3} - y^{5/3} = c^{5/3}$ (D) $x^{2/3} - y^{2/3} = c^{2/3}$

Question 30 (JAM 2017 • Q60 • ID: JAM_2017_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If

$$y(x) = \int_{\sqrt{x}}^x \frac{e^t}{t} dt, \quad x > 0$$

Question 31 (JAM 2017 • Q51 • ID: JAM_2017_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51 Let $y(x)$, $x > 0$ be the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0$$

satisfying the conditions $y(1) = 1$ and $y'(1) = 0$. Then the value of $e^2 y(e)$ is _____

Question 32 (JAM 2017 • Q41 • ID: JAM_2017_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 If the orthogonal trajectories of the family of ellipses $x^2 + 2y^2 = c_1$, $c_1 > 0$, are given by $y = c_2 x^\alpha$, $c_2 \in \mathbb{R}$, then $\alpha =$ _____

Question 33 (JAM 2017 • Q40 • ID: JAM_2017_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x \rightarrow \infty} y(x) = 0$. Then(A) $3k^2 + \ell < 0$ and $k > 0$ (B) $k^2 + \ell > 0$ and $k < 0$ (C) $k^2 - \ell \leq 0$ and $k > 0$ (D) $k^2 - \ell > 0$, $k > 0$ and $\ell > 0$

Question 34 (JAM 2017 • Q39 • ID: JAM_2017_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = (y - 1)(y - 3)$$

satisfying the condition $y(0) = 2$. Then which of the following is/are TRUE?

- (A) The function $y(x)$ is not bounded above
- (B) The function $y(x)$ is bounded
- (C) $\lim_{x \rightarrow +\infty} y(x) = 1$
- (D) $\lim_{x \rightarrow -\infty} y(x) = 3$

Question 35 (JAM 2017 • Q30 • ID: JAM_2017_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $y(x)$ be the solution of the differential equation

$$(xy + y + e^{-x})dx + (x + e^{-x})dy = 0$$

satisfying $y(0) = 1$. Then $y(-1)$ is equal to

- (A) $\frac{e}{e-1}$
- (B) $\frac{2e}{e-1}$
- (C) $\frac{e}{1-e}$
- (D) 0

Question 36 (JAM 2017 • Q29 • ID: JAM_2017_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 A particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^{2x} \sin x$$

is

- (A) $\frac{e^{2x}}{10}(3 \cos x - 2 \sin x)$
- (B) $-\frac{e^{2x}}{10}(3 \cos x - 2 \sin x)$
- (C) $-\frac{e^{2x}}{5}(2 \cos x + \sin x)$
- (D) $\frac{e^{2x}}{5}(2 \cos x - \sin x)$

Question 37 (JAM 2016 • Q56 • ID: JAM_2016_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 Let $y(x)$ be the solution of the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0, \quad y(0) = 1, \quad \left.\frac{dy}{dx}\right|_{x=0} = -1.$$

Then $y(x)$ attains its maximum value at $x =$ _____

Question 38 (JAM 2016 • Q47 • ID: JAM_2016_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 Let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be linearly independent solutions of the differential equation

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0.$$

If the Wronskian $W(y_1, y_2, y_3)$ is of the form ke^{bx} for some constant k , then the value of b is _____

Question 39 (JAM 2016 • Q35 • ID: JAM_2016_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Consider the differential equation

$$\sin 2x \frac{dy}{dx} = 2y + 2 \cos x, \quad y\left(\frac{\pi}{4}\right) = 1 - \sqrt{2}.$$

Then which of the following statement(s) is(are) TRUE?

- (A) The solution is unbounded when $x \rightarrow 0$
- (B) The solution is unbounded when $x \rightarrow \frac{\pi}{2}$
- (C) The solution is bounded when $x \rightarrow 0$
- (D) The solution is bounded when $x \rightarrow \frac{\pi}{2}$

Question 40 (JAM 2016 • Q22 • ID: JAM_2016_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 The general solution of the differential equation with constant coefficients

$$\frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

approaches zero as $x \rightarrow \infty$, if

- (A) b is negative and c is positive
- (B) b is positive and c is negative
- (C) both b and c are positive
- (D) both b and c are negative

MA

6/13

Question 41 (JAM 2016 • Q21 • ID: JAM_2016_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 Let $y(x)$ be the solution of the differential equation

$$\frac{d}{dx}\left(x \frac{dy}{dx}\right) = x; \quad y(1) = 0, \quad \frac{dy}{dx}\bigg|_{x=1} = 0.$$

Then $y(2)$ is

- (A) $\frac{3}{4} + \frac{1}{2} \ln 2$
- (B) $\frac{3}{4} - \frac{1}{2} \ln 2$
- (C) $\frac{3}{4} + \ln 2$
- (D) $\frac{3}{4} - \ln 2$

Question 42 (JAM 2016 • Q9 • ID: JAM_2016_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 One of the points which lies on the solution curve of the differential equation

$$(y - x)dx + (x + y)dy = 0,$$

with the given condition $y(0) = 1$, is

- (A) (1, -2) (B) (2, -1) (C) (2, 1) (D) (-1, 2)

Question 43 (JAM 2016 • Q8 • ID: JAM_2016_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 The nonzero value of n for which the differential equation

$$(3xy^2 + n^2x^2y)dx + (nx^3 + 3x^2y)dy = 0, \quad x \neq 0,$$

becomes exact is

- (A) -3 (B) -2 (C) 2 (D) 3

Question 44 (JAM 2015 • Q36 • ID: JAM_2015_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.6

The initial value problem

$$y' = \sqrt{y}, \quad y(0) = \alpha, \quad \alpha \geq 0$$

has

- (A) at least two solutions if $\alpha = 0$
(B) no solution if $\alpha > 0$
(C) at least one solution if $\alpha > 0$
(D) a unique solution if $\alpha = 0$

Question 45 (JAM 2015 • Q24 • ID: JAM_2015_Q24)

MCQ

+2 M

Neg: -0.67

Q.24

If $y(t)$ is a solution of the differential equation $y'' + 4y = 2e^t$, then

$$\lim_{t \rightarrow \infty} e^{-t} y(t)$$

is equal to

(A) $\frac{2}{3}$

(B) $\frac{2}{5}$

(C) $\frac{2}{7}$

(D) $\frac{2}{9}$

MA

9/16

Question 46 (JAM 2015 • Q12 • ID: JAM_2015_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 The orthogonal trajectories of the family of curves $y = C_1 x^3$ are

(A) $2x^2 + 3y^2 = C_2$

(B) $3x^2 + y^2 = C_2$

(C) $3x^2 + 2y^2 = C_2$

(D) $x^2 + 3y^2 = C_2$

Question 47 (JAM 2015 • Q9 • ID: JAM_2015_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 An integrating factor of the differential equation

$$\frac{dy}{dx} = \frac{2xy^2 + y}{x - 2y^3}$$

is

(A) $\frac{1}{y}$

(B) $\frac{1}{y^2}$

(C) y

(D) y^2

Question 48 (JAM 2015 • Q8 • ID: JAM_2015_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 If an integral curve of the differential equation $(y - x) \frac{dy}{dx} = 1$ passes through $(0, 0)$ and $(\alpha, 1)$, then α is equal to

(A) $2 - e^{-1}$

(B) $1 - e^{-1}$

(C) e^{-1}

(D) $1 + e$

Question 49 (JAM 2015 • Q2 • ID: JAM_2015_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let $y(x) = u(x) \sin x + v(x) \cos x$ be a solution of the differential equation $y'' + y = \sec x$. Then $u(x)$ is

(A) $\ln |\cos x| + C$

(B) $-x + C$

(C) $x + C$

(D) $\ln |\sec x| + C$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	10.8 to 10.9
2	NAT	+1	3
3	MCQ	+2	B
4	MCQ	+2	A
5	MCQ	+2	D
6	MCQ	+1	A
7	MCQ	+1	D
8	NAT	+2	3 to 3
9	NAT	+2	8 to 8
10	NAT	+1	7 to 7
11	NAT	+1	0.1 to 0.1
12	MSQ	+2	A;C
13	MCQ	+2	A
14	MCQ	+2	C
15	MCQ	+1	D
16	NAT	+2	1.12 to 1.25
17	NAT	+2	4 to 4

Q. No.	Type	Marks	Official Key
18	MCQ	+2	D
19	MCQ	+2	C
20	MCQ	+1	C
21	MCQ	+1	A
22	NAT	+2	0.5 to 0.5
23	NAT	+2	−2.80 to −2.70
24	NAT	+1	9 to 9
25	MSQ	+2	A;B;C
26	MCQ	+2	B
27	MCQ	+2	C
28	MCQ	+2	A
29	MCQ	+1	B
30	NAT	+2	1.34 to 1.36
31	NAT	+2	2.9 to 3.1
32	NAT	+1	1.9 to 2.1
33	MSQ	+2	C, D
34	MSQ	+2	B, C, D

Q. No.	Type	Marks	Official Key
35	MCQ	+2	B
36	MCQ	+2	C
37	NAT	+2	−0.3 to −0.25
38	NAT	+1	6.0 to 6.0
39	MSQ	+2	C;D
40	MCQ	+2	C
41	MCQ	+2	B
42	MCQ	+1	C
43	MCQ	+1	D
44	MSQ	+2	A;C
45	MCQ	+2	B
46	MCQ	+2	D
47	MCQ	+1	B
48	MCQ	+1	C
49	MCQ	+1	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (−1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.